LMEnt: A Suite for Analyzing Knowledge in Language Models from Pretraining Data to Representations

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Abstract

Language models (LMs) increasingly drive real-world applications that require world knowledge. However, the internal processes through which models turn data into representations of knowledge and beliefs about the world are poorly understood. To facilitate such studies, we present LMEnt, a suite including (1) a knowledge-rich pretraining corpus, fully annotated with entity mentions based on Wikipedia, (2) an entity-based retrieval method over pretraining data that outperforms existing tools by as much as 80.4%, and (3) 12 pretrained LMs with up to 1B parameters and 4K intermediate checkpoints, with comparable performance to popular open-source models on knowledge tasks. Together, these resources provide a controlled environment for analyzing connections between entity mentions in pretraining data and downstream performance. We show the utility of LMEnt by studying knowledge acquisition over training, finding that entity co-occurrence and mention forms—which are difficult to study with existing tools—affect learning trends. Moreover, as LMs form stronger associations between entities, their facts are harder to edit in-context, whereas inconsistencies in model predictions over training are indicative of editing success. We release LMEnt to support studies of knowledge in LMs, including knowledge representations, plasticity, editing, attribution, hallucinations, and learning dynamics.

🤗 huggingface.co/LMEnt
github.com/LMEnt

1 Introduction

Language models (LMs) capture substantial amounts of knowledge and beliefs about the world from their training data (Petroni et al., 2019; Roberts et al., 2020; AlKhamissi et al., 2022; Andreas, 2022). A fundamental, yet underexplored, question is how such knowledge representations are formed and shaped during pretraining. Namely, what is the interplay between data composition, training dynamics, and the internal knowledge mechanisms in LMs? An understanding of these processes could provide better control over the model's knowledge, potentially improving its factuality and reasoning.

A key prerequisite for studying this question is the ability to locate exactly where specific knowledge appears in the pretraining data. However, since such metadata is rarely available for pretraining corpora, many works that study knowledge acquisition have used synthetic data (Kassner et al., 2020; Hron et al., 2024; Chang et al., 2024; Allen-Zhu and Li, 2024; Zucchet et al., 2025; Ou et al., 2025), or post-hoc string-based search tools (Elazar et al., 2024; Liu et al., 2024a, 2025a) which lack robustness against variability in phrasing of semantically equivalent information. For example, when retrieving information about the American football team the *"Buffalo Bills"* (Fig. 2), these tools are less likely to retrieve documents that refer to the team as *"Buffalo"*, *"The Bills"*, or even *"the team"*, and suffer from noisy retrieval when queries are expanded with name aliases, as we observe in §5.2.

In this work, we introduce LMEnt (Language Models with annotated Entity mentions), a suite of pretrained models, a knowledge-rich pretraining corpus fully annotated with fine-grained entity mentions, and an entity-based retrieval method which allows full transparency into where a specific entity is mentioned in training. The focus on entity mentions stems from mounting evidence that entity names are central to the construction of knowledge representations in LMs (rather than individual tokens; Li et al., 2021; Meng et al., 2022; Geva et al., 2023; Gottesman and Geva, 2024).

Specifically, the LMEnt suite contains: **(1) A**

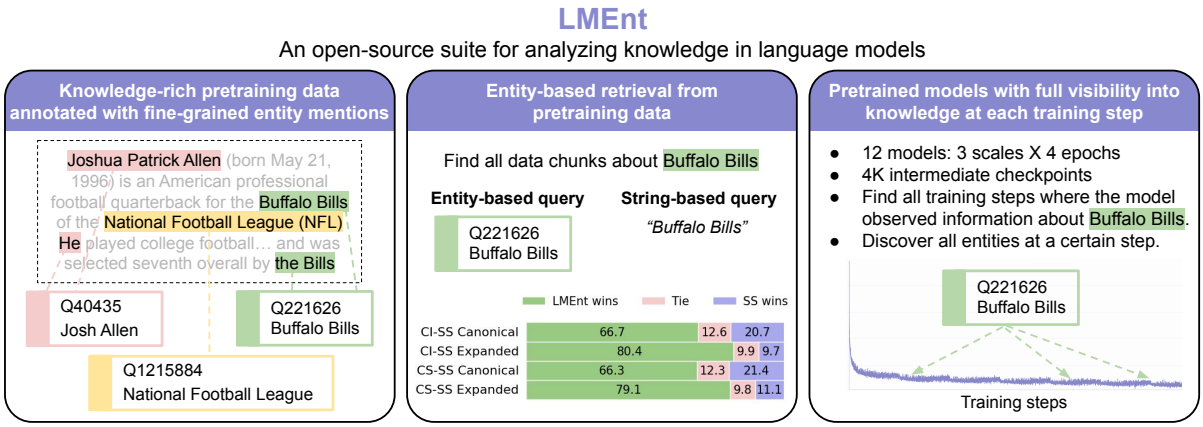


Figure 1: The LMEnt suite is composed of three components: (left) fine-grained entity mentions for every document in the pretraining corpus, (middle) an index that retrieves by the entity QID and outperforms string-based retrieval methods, (right) 12 models trained on 1, 2, 4, and 6 epochs where each step can be mapped to the entities it mentions, and each entity can be traced to the steps that introduce it.

pretraining corpus built on English Wikipedia with 7.3M fine-grained entity annotations across all documents, derived from three complementary sources: hyperlinks, entity linking, and coreference resolution (Fig. 1, left, §2). **(2) An Elasticsearch index with over 10.5M chunks**, enabling retrieval of all pretraining data chunks mentioning a specific entity by its unique Wikidata identifier (Fig. 1, center, §3). LMEnt's entity-based retrieval outperforms string-based methods similar to WIMBD (Elazar et al., 2024) and Infinigram (Liu et al., 2024a) on 66.3–80.4% of entities while maintaining >97% precision as retrieval depth grows, where string-based precision drops to as low as 27%. **(3) A collection of 12 LMs** with 170M, 600M, and 1B parameters, pretrained on the annotated data with 110 intermediate checkpoints per epoch and 4K checkpoints total (Fig. 1, right, §4). Despite only training LMEnt models on 0.03%–4.7% of the tokens used to train LMs of similar sizes, on factual question answering (Mallen et al., 2023), LMEnt models reach comparable performance (7.4% overall, 66% on questions about frequently co-occurring entities) to Pythia-1.4B (Biderman et al., 2023) (8.7%, 67%) and OLMo-1B (Groeneveld et al., 2024) (10.4%, 66%). They do have noticeably lower performance than OLMo-2-1B (OLMo et al., 2024) (13.9%, 74%) and SmolLM-1.7B (Allal et al., 2025) (14.9%, 73%) mostly due to recall failures on rare facts (§5).

Beyond releasing this resource, we use LMEnt to explore several open questions about how pretraining data shapes knowledge in LMs. In §6, we apply LMEnt to study knowledge acquisition and forgetting trends, and in-context editing. Because LMEnt reveals which training steps introduce chunks mentioning a given entity, we can examine how both fact frequency and location in training correlate with learning and forgetting. We find that LMs learn, and also forget, frequent facts more over the course of training, and that these trends differ across relation types (§6.1). Further, leveraging the diverse entity mention annotations, we find that LMs internalize facts better when entities are named explicitly in the data—an analysis infeasible with existing string-based tools (§6.2). Finally, we show that in-context editing success correlates with the frequency of entity co-occurrence in the data (§6.3) and to the rate of inconsistencies in the model's predictions over training (§6.4).

Overall, LMEnt is a novel tool for precisely tracking information during pretraining, laying the groundwork for studying the interplay between pretraining data and knowledge representations in LMs. We release LMEnt to the community as an open-source suite on HuggingFace, including the pretraining data, fine-grained entity mentions, pretrained models, and intermediate checkpoints. For ways LMEnt can support research into plasticity, factuality, hallucinations, and pretraining dataset composition, see §8.1.

2 Labeling the English Wikipedia with Fine-Grained Entity Mentions

To support analyzing knowledge evolution over training, we select a corpus rich in factual con-

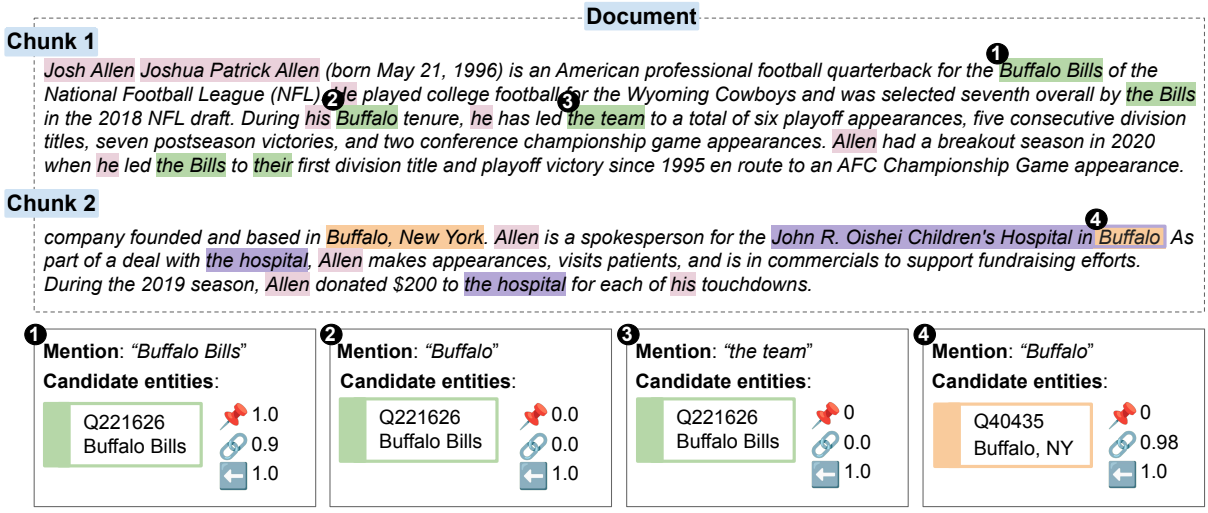


Figure 2: The document about *"Josh Allen"* is split into two chunks that are processed independently during pretraining. In Chunk 1, Q221626 is explicitly mentioned in Mention (1), found by hyperlinks and entity linking, and implicitly mentioned in Mention (3), found by coreference resolution. Although Q221626 (the football team) and Q40435 (the city of Buffalo) share the surface form "*Buffalo*," LMEnt correctly disambiguates them: Mention (2) is resolved to Q221626 with 1.0 confidence through coreference resolution, whereas Mention (4) is linked to Q40435 with 0.98 confidence using entity linking.

tent and annotate its documents with fine-grained entity mentions (Fig. 1, left). These annotations enable tracking exactly which entities a model observes at each training step when training on this data. Next, we explain our choice of Wikipedia for pretraining data, and the annotation process.

Pretraining data Wikipedia is a natural choice for the source of pretraining data because it is a knowledge base that is structured around entities, such as people, locations, and world events. Importantly, there is an integrated system for disambiguating entities using unique identifiers, called *Wikidata QIDs* (Vrandečić and Krötzsch, 2014), and related entity pages are connected through hyperlinks. Moreover, it provides a snapshot of world knowledge at a given time which minimizes contradicting information. Wikipedia is also commonly used to build knowledge benchmarks (e.g., Petroni et al., 2021; Sciavolino et al., 2021; Lewis et al., 2021; Mallen et al., 2023), making it well-suited for tracking how changes in pretraining affect learning.

Annotation objectives Our goal is to be able to query which entities a model saw at any given training step. To achieve this, we built an annotation process that maps character spans in the pretraining data which mention an entity, called *entity mentions*, to unique Wikidata QIDs.[1] Entity mentions are created for each data chunk, but are not exposed to the model during training—rather, the QIDs are the keys used to retrieve all the data associated with an entity for post-hoc analysis. We design entity mentions to satisfy the following criteria (see Fig. 2 for illustration):

C1: Disambiguate between entities with similar names and aliases, e.g., *"Buffalo"* for the football team and *"Buffalo"* for the city.

C2: Capture entities referenced indirectly through descriptive phrases and pronouns, e.g., *"the team"*, *"their"* and *"the hospital"*.

C3: Resolve entity mentions using the full document context e.g., *"Allen"* (Chunk 2) should be annotated with reference to his explicit name, *"Joshua Allen"* (Chunk 1).

We meet **C1** and **C2** by leveraging three complementary sources for annotating entity mentions (§2.1), and **C3** by applying the scoring process to the entire document before chunking (§2.2).

2.1 Entity Mention Sources

Hyperlinks identify the first occurrences of explicit entity mentions like Mention (1) in Fig. 2

[1]The annotation pipeline used 8 H100 GPUs (80GB VRAM each). Running coreference resolution and entity linking took 5 days with full GPU utilization.

which links to the URL of the football team's Wikipedia page, and can be mapped to Q221626 using Wikidata's Query Service. To extract hyperlinks, we parse the raw XML Wikipedia dump and extract the `url` attributes from `href` tags.

Entity linking is used to increase coverage of direct mentions like *"Buffalo"* since hyperlinks offer limited coverage within a document.[2] We use ReFinED (Ayoola et al., 2022), a state-of-the-art modular entity linking system that performs mention detection, fine-grained typing, and entity disambiguation. ReFinED supports zero-shot entity linking by encoding the mention context and candidate entity descriptions using RoBERTa (Liu et al., 2019), and selecting the entity whose description embedding best aligns with the mention representation. By replacing the underlying base of entity descriptions from our Wikipedia dump, we can identify mentions of new entities and map them to their QIDs without retraining ReFinED.

Coreference resolution connects indirect mentions, like pronouns, aliases, and generic descriptors, to their entity QIDs. The two sources above are used to extract explicit entity mentions; however, entities may still be referred to implicitly, e.g., *"the team"* in Fig. 2. To fill in this gap, we use the Maverick coreference resolution model (Martinelli et al., 2024), which achieves state-of-the-art performance on WikiCoref (Ghaddar and Langlais, 2016). To run Maverick at scale, we reduce its memory footprint by replacing the model backbone, allowing us to parallelize inference on documents (see §B.1 for details). Maverick outputs clusters of mentions that refer to the same entity. However, these mentions must still be mapped to their QIDs, which we discuss below.

2.2 Scoring Entity Mentions in a Document

Every source yields a set of entity mentions per document in the corpus, and each entity mention is defined by its character span $m = (c^{\text{start}}, c^{\text{end}})$. To consolidate the mentions found across sources, we define source-specific scoring procedures which quantify how confidently a mention can be mapped to a QID (Fig. 2):

- **Hyperlinks (H)** 📌 This is the most reliable source since hyperlinks are manually added by human editors. We define $\mathbf{H}(m, \texttt{Q221626}) = 1$ if there is a hyperlink spanned by m directing to the Wikipedia page associated with Q221626, and $\mathbf{H}(m, \texttt{Q221626}) = 0$ otherwise.
- **Entity linking (EL)** 🔗 The ReFinED model already performs entity disambiguation and provides a score reflecting its confidence that a mention links to a QID, e.g., $M(m, \texttt{Q40435}) = 0.98$. Therefore, we simply use this score such that $\mathbf{EL}(m, \texttt{Q40435}) = M(m, \texttt{Q40435})$.
- **Coref (C)** ⬅️ Suppose we are trying to resolve the QID of the implicit mention, *"the team"*. We run Maverick on the document, and it finds that *"the team"* is related to the cluster of mentions containing *"Buffalo Bills"*, *"the Buffalo"*, *"the Bills"*, and *"their"*. How can we leverage the cluster to identify the QID of *"the team"*? We compute a distribution of scores over all QIDs mapped to some mention in the cluster by the other two sources. In this case, Q221626 is the only QID supported by this cluster so $\mathbf{C}(\text{"the team"}, \texttt{Q221626}) = 1.0$. Since this score is computed using the entire cluster, the **C** score is the same across all mentions in the cluster. Occasionally, a coreference mention encapsulates multiple entities, as in *"John R. Oishei Children's Hospital Buffalo"*, which leads to a score distribution over several QIDs. In this case of mapping ambiguity, we rely on textual similarity of mentions in the cluster to promote one QID over another. §B.2 provides the formal definition of this score, and a detailed explanation of how we resolve mapping ambiguity.

The final entity mention structure maps each mention m to a list of candidate QIDs with up to three source scores per candidate. This list structure accommodates spans that encompass multiple entities, and multiple scores per candidate allow us to filter chunk retrieval based on source type and confidence thresholds (§3.2). §B.3 includes a qualitative analysis of error patterns over a random sample of 112 mentions, justifying the design of the scoring procedures.

3 Retrieval from Pretraining Data

Now that we have entity mentions for every document (Fig. 1, left), we need to process the docu-

[2] Wikipedia style guidelines recommend linking only the first mention of an entity in a document: https://en.wikipedia.org/wiki/Wikipedia:Manual_of_Style/Linking.

ments into chunks of information for pretraining, and associate the relevant entity mentions with each chunk. By iterating over the entity mentions of each chunk at a particular training step, we can track the entities seen over training (Fig. 1, right). §3.1 describes how documents are processed into chunks, and §3.2 discusses entity-based retrieval over our index of chunks (Fig. 1, center).

3.1 Data Processing for Training

Tokenization Documents are tokenized with the dolma2-tokenizer (Soldaini et al., 2024), resulting in a dataset with 3.6 billion unique tokens. For reference, this corresponds to around 0.09% of the tokens used to train OLMo-2 (OLMo et al., 2024).

Chunking A chunk is a sequence of tokens that the model independently processes per training step. Unlike existing chunking strategies, like *concat-and-chunk* used in Liu et al. (2019); Ott et al. (2019); Touvron et al. (2023a,b), we ensure that each chunk only contains content from a single document using the Variable Sequence Length Curriculum (Pouransari et al., 2024). This prevents the model from learning spurious correlations by attending to unrelated documents.

Chunking curricula split documents at some sequence length, which can split entity mentions across chunk boundaries. To prevent this, we modify the chunking logic to terminate just before a mention and pad the remaining tokens to the desired length. The entity mentions for a chunk are defined as those fully contained within it, and we adjust them so they are relative to the chunk's boundaries. Practically, we extend the OLMo-Core framework (OLMo et al., 2024) to include entity mentions in the chunk metadata.

3.2 Entity-Based Chunk Retrieval

We build an Elasticsearch index (Banon, 2010) containing one entry per chunk in the pretraining corpus. Each entry stores the `chunk_id`, the chunk text, and its associated entity mentions. As illustrated in Fig. 2 and described in §2.2, each entity mention is linked to a set of candidate entities, where each candidate is identified by its QID and accompanied by three source-specific scores.

To **retrieve all chunks** that mention a target entity, we query the index for chunks containing at least one mention whose candidate list includes the target entity's QID (see §2.2). Retrieval can also be filtered by source type and score thresholds. Retrieval algorithms typically apply an ordering scheme to the items they return (Robertson et al., 1994). To accommodate this, we assign a score to each chunk based on the entity mentions that match the retrieval query. For every matching mention, we compute a weighted average of its three source-specific scores. The chunk's final ranking score is defined as the maximum over the average scores of all its matching mentions.

To **retrieve all training steps** that mention a target entity, we first retrieve all corresponding chunks from the index, and then map each retrieved chunk to the training step in which it was introduced. This mapping is straightforward since the `chunk_id` is the index of the chunk in the dataset, and its associated training step is the index of the batch that contains it in the dataloader.[3]

4 The LMEnt Suite

With the procedures for annotating entity mentions and processing pretraining data established, we now provide an overview of the resulting pretraining data index and `LMEnt` models.

Pretraining data index Our pretraining data includes 3.6B tokens over 10.5M chunks, and features more than 7M different entities. The 400M mentions extracted are composed of 115M mentions from hyperlinks, 203M from entity linking, and 310M from coreference resolution. Additional statistics are provided in §A.1, Tab. 2.

Pretrained models We introduce a collection of models trained on the dataset, which serve as a testbed for analyzing knowledge evolution over training. Specifically, we train models of three different sizes, 170M, 600M, and 1B parameters, based on the OLMo-2 architecture (OLMo et al., 2024). Notably, the 170M model has the compute-optimal size for 3.6B tokens (Hoffmann et al., 2022). Each model is trained for 1, 2, 4, and 6 epochs (3.6B, 7.2B, 14.4B, 21.6B tokens, respectively), resulting in four variants per size. Each epoch initializes a different batching and ordering of the chunks. We also release intermediate checkpoints every 1,000 training steps, yielding 110 checkpoints per epoch and proportionally more for longer training schedules. Additional details on model training are in §B.4.

[3] There are multiple chunks per batch, determined by a global batch size of 32,648 tokens.

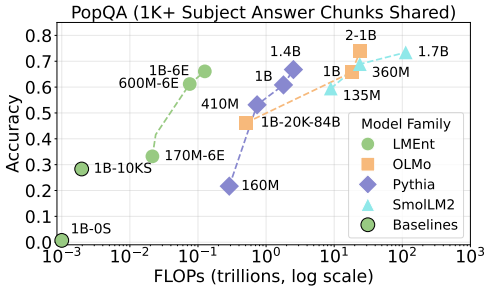


Figure 3: Accuracy on PopQA questions, whose subject and answer entities co-occur frequently, as a function of compute budget. LMEnt models achieve comparable performance to other models with better compute efficiency.

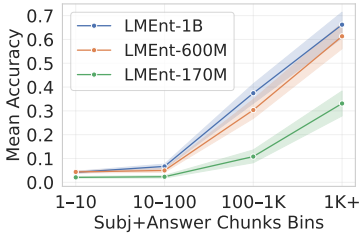


Figure 4: Mean accuracy on PopQA questions binned according to the number of chunks that the subject and answer entities co-occur in. Increasing model size helps learning associations between entities that appear more frequently together.

5 Experiments

In the previous sections, we described the construction and composition of the LMEnt suite. Here, we establish the credibility of the suite by showing that LMEnt models match the knowledge recall performance of existing open-source models (§5.1), and that entity-based chunk retrieval using LMEnt entity mentions outperforms string-based methods (§5.2).

5.1 Model Performance

Experimental setup To evaluate the LMEnt models, we use benchmarks that test for knowledge in Wikipedia. We choose two widely adopted benchmarks: PopQA (Mallen et al., 2023) and PAQ (Lewis et al., 2021). PopQA contains questions about 11K entities with varying popularities, and PAQ has 65M questions generated from Wikipedia passages. To keep computational costs manageable, we subsample PAQ and only consider samples containing the same subject entities found in PopQA. This results in an evaluation set of approximately 70K questions.[4] Results on PAQ are found in §A.2. In addition to knowledge-centric tasks, we report results on commonsense reasoning, multiple choice, and reading comprehension tasks in §A.4. Since LMEnt models are not instruction-tuned, we convert all samples into cloze-style prompts, such that the answer is the next expected phrase. For example, the original QA-style query "*What TV network does Funny or Die Presents air on?*" translates to "*Funny or Die Presents airs on the TV network*". Full details of this conversion process are in §B.6.

[4]Evaluating a 1B parameter model on all PAQ questions using one H-100 80GB GPU would take 200 days.

Baselines We evaluate the performance of LMEnt models against leading open-source models of comparable sizes, including Pythia (160M, 410M, 1B, 1.4B) (Biderman et al., 2023), OLMo (1B, 1-20K-84B) (Groeneveld et al., 2024), OLMo-2-1B (OLMo et al., 2024), and SmolLM2 (135M, 360M, 1.7B) (Allal et al., 2025). We include OLMo-1-20K-84B—an intermediate checkpoint of OLMo-1B trained for 20K steps on 84B unique tokens—because it is the most comparable baseline to our models in terms of training token count.[5] We also include two intermediate LMEnt model baselines: LMEnt-1B-0E (randomly initialized) and LMEnt-1B-0.1E (trained for 10K steps). These serve to illustrate how more training data improves knowledge recall capabilities.

LMEnt models are valid resources for studying knowledge Fig. 3 shows the performance of LMEnt models and the baselines as a function of compute budget, measured in FLOPs. Despite being trained with two orders-of-magnitude less compute, LMEnt models achieve performance comparable to both Pythia and OLMo-1B on questions in PopQA whose subject and answer entities co-occur frequently in the pretraining dataset. We also present results for all frequency levels in §A.2 which show similar trends. However, since PopQA is dominated by questions about tail facts, overall accuracy is lower across all models.

Notably, the LMEnt-600M model surpasses the OLMo-1B intermediate checkpoint and the Pythia-1B checkpoint, which reaffirms that pretraining on Wikipedia enables models to acquire substantial factual knowledge (Devlin et al., 2019)

[5]Notably, OLMo-1-20K-84B is trained on 4 times more data than our largest LMEnt-1B model, and it may not have observed all the facts we evaluate on.

	Retrieval	Match Examples
LMEnt	QID + Scores	Buffalo, New York; Buffalo; the Queen City; this city; the City's; the City of Buffalo; the city Buffalo; a town on the banks of Lake Erie; his hometown
CS-SS	Canonical	Buffalo, New York
	Extended	Buffalo, New York; Buffalo, NY; Buffalo, N. Y.; City of Light; The Queen City; The Nickel City
CI-SS	Canonical	buffalo, new york
	Extended	buffalo, new york; buffalo, ny; buffalo, n. y.; city of light; the queen city; the nickel city

Table 1: Matched entity mentions of "*Buffalo, New York*" for different retrieval methods. Entity-based retrieval over the LMEnt index matches on Q40435. The Canonical method matches only the entity's canonical name, whereas the Extended method matches any of its aliases.

and achieve better compute-performance trade-offs. Though this evaluation aligns with the pretraining objective of LMEnt models, while the baseline models are trained on a broader mixture of web, math, and code data, it supports their validity as resources for studying knowledge.

Measuring the effect of model scale on knowledge encoding Since all LMEnt models are pretrained on the same data, we can isolate the effect of model size on knowledge acquisition (Roberts et al., 2025). Fig. 4 shows that increasing model size improves learning of frequent facts (subject and answer co-occur in ≥ 100 chunks), while having little effect on tail facts (1–100 chunks). We leave exploring how scaling model size further can enhance knowledge learning for future work.

5.2 Chunk Retrieval Performance

A key use-case of the LMEnt index is retrieving all the pretraining data chunks that mention a specific entity. Here, we compare the quality of entity-based retrieval (§3.2) to existing string-based retrieval methods.

Experimental setup We evaluate on a test set of 1K entities, chosen via stratified sampling by their number of hyperlinks. We use the entity-based retrieval over the LMEnt index to retrieve their corresponding chunks, matching both on entity QID and on at least one of the score thresholds: $\mathbf{H} = 1$, $\mathbf{EL} \geq 0.6$, $\mathbf{C} \geq 0.6$. These thresholds were empirically determined using a development set of 60 entities, described in §A.5. We then compare the

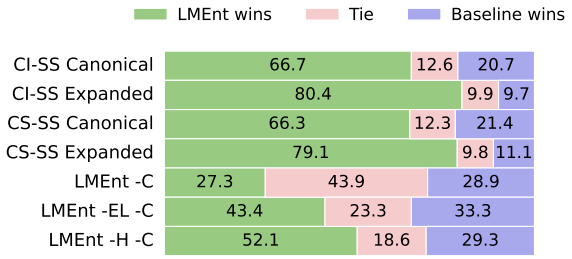


Figure 5: Pairwise wins rate for entity-based retrieval (LMEnt) with multiple string-based methods and ablations. LMEnt's entity-based retrieval outperforms string-based methods by 66.7%–80.4%. Ablations (bottom three rows) show that hyperlinks and entity linking are the most crucial sources of entity mentions.

set of retrieved chunks to those retrieved by popular string-based methods.

To evaluate retrieval precision, we use Gemini 2.5 Flash (Comanici et al., 2025) as a judge (Gu et al., 2024; Li et al., 2025) which predicts "Yes" or "No" based on whether the mention directly relates to the target entity. The prompt used and the statistical test justifying our use of an LLM-judge are provided in §B.5. Since some entities are mentioned very frequently (Fig. 16), we randomly sample 100 chunks from the set retrieved per method and entity, and measure the absolute number of chunks judged as "Yes" for each entity.

Baselines We compare to string-based baselines that retrieve chunks using either case-sensitive (CS-SS) or case-insensitive (CI-SS) exact matches of entity names within the chunk text. The Canonical variant matches only on the entity's canonical name, and the Extended variant additionally matches against the entity's Wikidata aliases which is a common strategy used to improve recall (e.g., Cohen et al., 2024; Yang et al., 2024).

The CS-SS Canonical baseline approximates the Infinigram (Liu et al., 2024a) tool which matches based on case-sensitive n-grams, and CI-SS Canonical baseline approximates WIMBD (Elazar et al., 2024) which retrieves on exact case-insensitive string matches. Although CS-SS Canonical relies on exact string matches rather than n-gram matches, it is a stronger baseline since shorter references introduce noise due to alias ambiguity (e.g., the n-gram "*Buffalo*" indiscriminately matches chunks that refer to the city, animal, or football

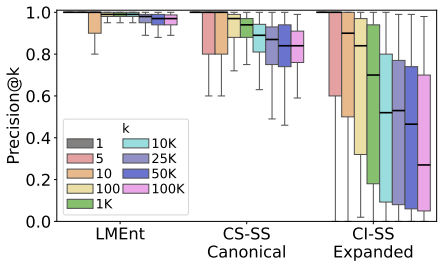


Figure 6: Precision at various retrieval depths (k), the top-k chunks retrieved per entity. As depth increases, LMEnt maintains $>$ 97% precision, while CS-SS Canonical and CI-SS Expanded decrease to 84% and 27%.

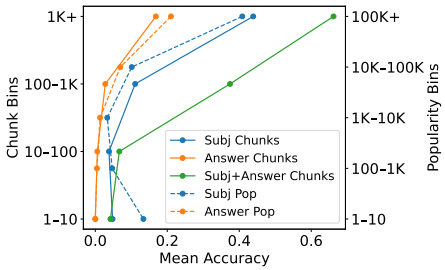


Figure 7: Accuracy for LMEnt-1B on PopQA. Subj+Answer Chunks counts chunks that mention both the subject and answer entities of a question. Answer Chunks and Subject Chunks count chunks that mention the answer and subject entities individually. Subj Pop and Answer Pop are pageview popularities from Mallen et al. (2023). Subj+Answer Chunks correlates best with the model's performance.

team). Tab. 1 summarizes the retrieval strategies used by LMEnt and the baselines. For clarity, we also include examples of mentions that triggered the retrieval of chunks for Q40435.

Entity-based retrieval largely outperforms the baselines Fig. 5 presents pairwise win rates between retrieval methods, where a win is defined as one method retrieving more "Yes"-judged chunks than another for a given entity. Entity-based retrieval (LMEntwins) consistently outperforms all other methods, retrieving more correct mentions for 66.3%–80.4% of the entities. Surprisingly, the Expanded variants of both CI-SS and CS-SS perform worse than their canonical counterparts by 40%, suggesting that additional noise introduced by alias expansion outweighs any gains in recall (§A.6, Fig. 15). Additionally, we observe that entity-based retrieval is superior to string-based methods across all entity frequency levels—notably on tail and torso entities which together make up 99.7% of the entities in Wikipedia—and often by a substantial number of chunks (§A.7, Fig. 16; §A.8, Fig. 17).

Ablating entity mention sources reduces retrieval quality We also ablate sources of entity mentions in entity-based retrieval and examine the impact on win rate (Fig. 5 last three rows). Hyperlinks are the most valuable component as relying solely on them (LMEnt -EL -C) wins on 33.3% of entities. In contrast, using only entity linking (LMEnt -H -C) lowers wins to 29.3%. This indicates that while hyperlinks play a crucial role, entity linking alone remains reasonably effective—highlighting the potential of applying LMEnt's annotation pipeline to pretraining corpora beyond Wikipedia, where hyperlinks are unavailable. Removing coreference resolution (LMEnt -C) has little effect, because judging chunks with only implicit mentions is difficult. For example, when Donald Trump is referred to solely as "*his Mexico City Policy*", it is difficult to identify that "*his*" implicitly refers to Trump without the context earlier in the document which mentions him explicitly.

LMEnt maintains high precision as the number of retrieved chunks increases We also analyze the precision of retrieved chunks at varying retrieval depths, k. For each method, we consider only the top-k chunks retrieved per entity, and randomly sample 100 chunks if $k \geq 100$, for feasibility. An LLM then judges the mention, and precision is computed as the fraction of "Yes" responses out of k. As shown in Fig. 6, LMEnt's entity-based retrieval maintains high precision $\geq$ 97% as k increases, while precision substantially declines for all other methods, reaching 27% and 84% for CI-SS and CS-SS, respectively. This indicates that as more chunks are retrieved, string-based methods introduce noise while LMEnt maintains high quality.

Entity co-occurrence better aligns with model performance than popularity The number of Wikipedia pageviews for a subject entity, denoted as Subj Pop, has been the standard proxy for real-world popularity and is known to correlate with a model's ability to recall facts about it (Mallen et al., 2023). In Fig. 7, we compare model performance against several potential indicators.

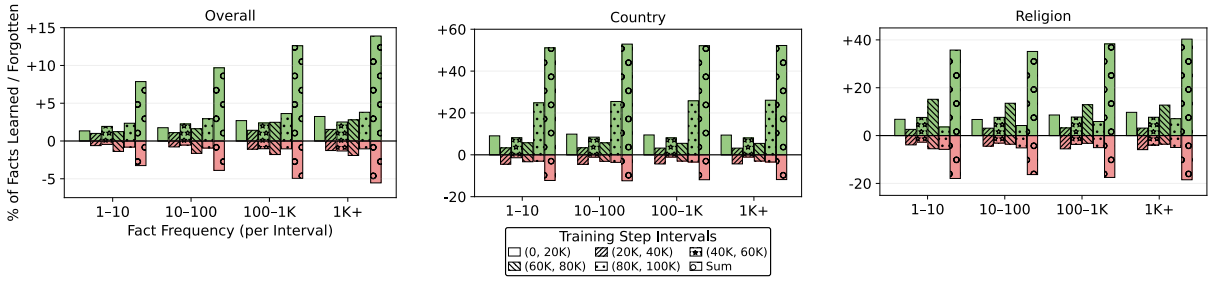


Figure 8: Percentage of facts learned and forgotten by fact frequency between intermediate checkpoints of LMEnt-1B. Overall (left) shows that while fact frequency correlates with knowledge gains, both learning and forgetting increase with frequency. The learning and forgetting trends for specific Country (middle) and Religion (right) relations differ across training step intervals. Overall learning is lower since the model struggles to answer questions about certain relations (e.g., Screenwriter in Fig. 18).

In PopQA, each question concerns a (Subject, Answer) entity pair. Leveraging LMEnt's entity-based retrieval, which can retrieve chunks containing many co-occurring entities, we introduce a new indicator: Subject+Answer Chunks. This is defined as the number of chunks in which the subject and answer entities co-occur. This co-occurrence shows the strongest correlation with model performance. Interestingly, this trend is also consistent for models trained on pretraining data that extend beyond Wikipedia (§A.3, Fig. 14).

6 Applications of LMEnt

We demonstrate the utility of LMEnt by focusing on two fundamental applications: knowledge acquisition and in-context knowledge editing. Specifically, we study learning and forgetting trends over training (§6.1), how entity mention forms relate to learning (§6.2), and how in-context knowledge editing relates to entity co-occurrence frequency (§6.3) and inconsistencies in model predictions over training (§6.4).

6.1 Knowledge Acquisition in Pretraining

We study at which point during training models acquire factual knowledge and how this relates to fact frequency, which has been previously approximated using document frequency (Kandpal et al., 2023), term co-occurrence (Elazar et al., 2022; Merullo et al., 2025) and n-gram (Wang et al., 2025) frequencies. In contrast to prior work, we use LMEnt entity mentions, which provide more accurate estimates of fact frequency compared to string-based methods.

Experiment We evaluate the LMEnt-1B model every 20K steps on PopQA. We define a fact by a (Subject, Answer) entity pair, which has been shown to be an effective fact proxy (Elsahar et al., 2018). A fact is considered "learned" if the model correctly answers at least one question linking the subject to the answer. Between training steps $(i, i{+}20\text{K})$, we consider all the facts seen in the training step interval and measure (a) *fact frequency*: the number of chunks in this interval where the subject and answer co-occurred, (b) *% of learned facts*: facts that were not learned at step i and then learned at step $i{+}20$K, and (c) *% of forgotten facts*: facts that were learned at step i and then not learned, i.e. "forgotten", at step $i{+}20$K.

Findings Fig. 8 (left) shows the percentage of facts learned and forgotten as a function of both fact frequency and the training step interval in which the fact was introduced. Focusing on trends across fact frequencies (left, Sum), the model recalls frequent facts from memory more effectively, which is consistent with prior works (Elazar et al., 2022; Yu et al., 2023; Liu et al., 2025b). However, we also see that LMs forget very frequent facts (1K+, -5.2%), and successfully learn rare facts (1-10, +8%). This implies that fact frequency alone does not dictate learning and forgetting dynamics, and may be confounded by other factors like when a fact is introduced in training (discussed next), or its entity mention forms (see §6.2).

To analyze how the timing of a fact's introduction affects its acquisition, consider any fact frequency bucket in Fig. 8 (left). We see that the model acquires knowledge in the first 20K training steps, and continues to learn and forget facts as training proceeds, with the highest percentage of facts learned in the last 20K training steps. Moreover, knowledge acquisition is not correlated with the cosine learning rate schedule, which decreases

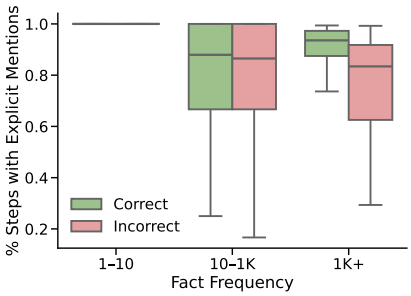


Figure 9: LMEnt-1B is more accurate when entities co-occur in their explicit forms, implying that entity mention forms influence learning.

after 1K warmup steps, as acquisition is highest in the last training step interval where the learning rate is lowest.

Further, we see varied learning dynamics when restricting this analysis to specific relations: Country (middle) and Religion (right). For example, the training step interval with highest learning is (80K, 100K) for country, but (60K, 80K) for religion. This observation aligns with previous findings that knowledge recall trends over training differ across relations and entity types (Yang et al., 2024; Balesni et al., 2024). Results for additional relations can be found in §A.9.

6.2 Entity Mention Forms and Learning

The previous experiment shows that fact frequency alone does not fully explain learning and forgetting trends. So far we treated all co-occurrences of the subject and answer entities as equally informative, but entities can be mentioned with various entity mention forms, e.g., "*Donald Trump*" vs. "*the 47th president*". Here, we leverage the fine-grained LMEnt entity mentions to examine the relation between how an entity is mentioned in the pretraining data and the model's ability to learn facts about it. Notably, such analysis is infeasible with string-based methods since they suffer from alias ambiguity.

Experiment We restrict our analysis to chunks in which a (Subject, Answer) entity pair co-occur, and determine whether both the subject and answer entities are mentioned explicitly (e.g., canonical names like "Donald Trump") or implicitly through aliases (e.g., "*the 47th president*") or pronouns (e.g., "*he*"). For each (Subject, Answer) entity pair, we then compute the proportion of training steps that introduce chunks which explicitly mention both entities.

Findings Fig. 9 shows the percentage of training steps that mention both the subject and answer entities explicitly, grouped by fact frequency and correctness. For rare facts (1–10), which appear in more focused contexts, all of the training steps contain explicit mentions regardless of correctness, indicating that a different factor is at play (e.g., location in training). For more frequent facts (10-1K), correctly answered questions are associated with a slightly higher proportion of steps with explicit mentions, and this gap is more pronounced for highly frequent facts (1K+). This suggests that mention forms affect how models learn factual knowledge, and may contribute to why LMs forget frequent facts. We encourage future work using LMEnt to systematically study these effects, including causal interventions that replace implicit mentions with explicit ones to isolate impacts on learning and forgetting.

6.3 Entity Co-occurrence and In-Context Editing Success

We saw that the co-occurrence of subject and answer entities correlates with the model's performance on related queries (§6.1). Here, we extend this analysis to study how co-occurrence statistics affect the model's ability to edit its learned knowledge in-context (Zheng et al., 2023; Yu et al., 2023; Cohen et al., 2024).

Experiment We evaluate checkpoints of the LMEnt-1B model trained for six epochs that were saved every 100K training steps. Focusing on the questions in PopQA that were answered correctly by at least one of these checkpoints, we generate up to 50 counterfactual answers per question by retrieving all the chunks that mention the subject entity and choosing counterfactual entities that (i) match the original answer type and (ii) co-occur with the subject in the pretraining data. For example, for the fact "*The capital of France is Paris*", the city "*Rome*", which also co-occurs with "*France*" in training, can serve as a counterfactual answer. To evaluate in-context knowledge editing, we prompt every checkpoint with the edited fact followed by the original cloze-style prompt, e.g., "*The capital of France is Rome. The capital of France is*", and consider editing successful if the LM outputs the counterfactual answer e.g., "*Rome*".

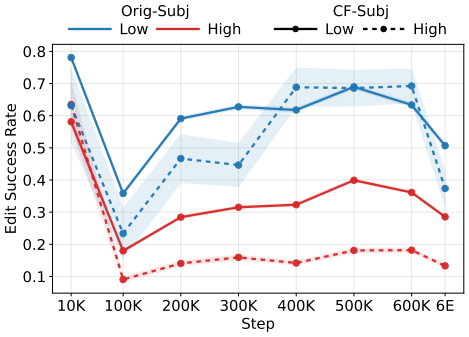


Figure 10: In-context editing success across training checkpoints, split by the co-occurrence frequencies of `Orig-Subj` and `CF-Subj` pairs. LMs are more rigid to editing facts with strong associations with either pair (red $<$ blue, dash $<$ solid). `6E` refers to the final model at epoch six.

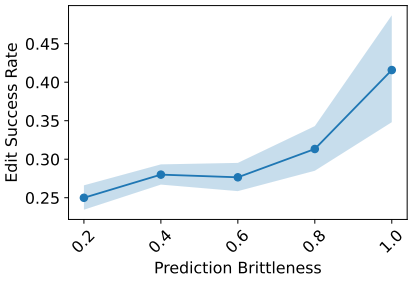


Figure 11: In-context editing success of the `LMEnt`-1B model increases with prediction brittleness. Logistic regression found a monotonic relationship ($\beta = 0.54, z = 4.97, p = 6.65e^{-07}$).

Findings Fig. 10 shows in-context editing success over checkpoints, split by the co-occurrence frequencies of two entity pairs: `Orig-Subj`, the (Subject, Original Answer) entity pair like ("*France*", "*Paris*"), and `CF-Subj`, the (Subject, CF Answer) entity pair like ("*France*", "*Rome*"). Each pair is labeled low-frequency (1–10 training steps) or high-frequency (100+), and an edit is categorized by both its `Orig-Subj` and `CF-Subj` frequency labels, e.g., the dashed blue line shows edits with low `Orig-Subj` and high `CF-Subj`.

We find that `Orig-Subj` co-occurrence plays the dominant role—when the original answer and subject are weakly associated (blue), in-context editing success is highest. `CF-Subj` co-occurrence also matters—when the counterfactual entity and subject are weakly associated (solid), the LM more readily adopts the new relation. Examining trends over training, we see high edit success (58%-78%) for the first checkpoint (10K training steps). After approximately one epoch, the model's knowledge becomes rigid for its strong associations (red/dashed). However, for facts with weaker associations, the model exhibits a non-monotonic pattern: editability decreases after the first epoch, partially recovers mid-training, and then declines again at the end. This behavior aligns with prior work showing that continual learning is more effective from intermediate checkpoints (Springer et al., 2025).

6.4 Model Brittleness and In-Context Editing

Here, we examine how inconsistencies in the model's predictions across training relate to the model's ability to edit its knowledge in-context.

Experiment We evaluate the final checkpoint using the same dataset as in §6.3. We define *prediction brittleness* as the frequency of "correctness flips" (Correct→Incorrect and Incorrect→Correct) across six checkpoints, sampled every 100K training step, with steps beginning from the first third of training to exclude noisy early-stage prediction fluctuations. Examples whose correctness never changes are omitted as they do not exhibit prediction brittleness.

Findings Fig. 11 shows that in-context editing success increases with prediction brittleness: facts whose answers fluctuate more during training are easier to edit. A binomial logistic regression analysis confirms the relation is monotonic and positive. This result highlights a connection between training dynamics and the editability of the final model, motivating future work on what drives such brittleness and the relationship between knowledge stability and model safety.

7 Related Work

Analysis of data acquisition in LMs Prior work on how pretraining data shapes factual knowledge has largely relied on synthetic setups, due to limited metadata available for pretraining corpora. These include injecting synthetic facts (Chang et al., 2024), constructing artificial biographies (Allen-Zhu and Li, 2024; Zucchet et al., 2025), pretraining on structured fact triplets (Kassner et al., 2020; Hron et al., 2024), and identifying knowledge circuits using synthetic data (Ou et al., 2025). In contrast, `LMEnt` enables controlled investigation in realistic settings using natural data.

Other lines of work examine the influence of individual training examples on learning (Han et al., 2020; Hammoudeh and Lowd, 2024), localize where knowledge is encoded in model parameters (Kim et al., 2025), and analyze which properties of pretraining data give rise to hallucinations (Sun et al., 2026; Zucchet et al., 2025). These approaches can benefit from LMEnt, which allows direct comparison of model checkpoints before and after seeing specific facts.

Retrieval over pretraining corpora LMEnt's entity-based retrieval outperformed string-based retrieval methods (§5.2) which rely on exact matches (Elazar et al., 2024; Liu et al., 2025a) or n-gram matches (Liu et al., 2024a). Many works that study the connection between pretraining data statistics and knowledge acquisition (Razeghi et al., 2022; Kandpal et al., 2023; Elazar et al., 2022; Kang and Choi, 2023; Merullo et al., 2025; Wang et al., 2025) can use LMEnt's entity-based retrieval to refine their findings.

Open-source LM toolkits Similar to LMEnt, Pythia (Biderman et al., 2023), OLMo (Groeneveld et al., 2024; OLMo et al., 2024), LLM360 (Liu et al., 2024b), SmolLM (Allal et al., 2025; Bakouch et al., 2025), and NVIDIA (Kuchaiev et al., 2019; Basant et al., 2025) released a suite of models along with their training data and training framework to facilitate the study of scaling laws and training dynamics in relation to the pretraining data. Other works train their own collection of models to study scaling laws for data curation (Magnusson et al., 2025) and specific downstream tasks (Bhagia et al., 2025). LMEnt differs from prior work by providing a practical toolkit for studying factual knowledge in LMs, with a particular emphasis on easily attributing when and where specific information appears in training.

8 Conclusion and Discussion

We introduce LMEnt, a suite of 12 LMs, pretraining data fully annotated with entity mentions, and an entity-based retrieval index that facilitates studying the connection between knowledge acquisition and pretraining data. We show that LMEnt models are capable of knowledge recall tasks, and that our entity-based retrieval outperforms string-based methods on 66.3%–80.4% of entities. Overall, LMEnt serves as a flexible and extensible testbed for investigating a broad set of questions regarding knowledge and representations in LMs. In the following sections, we list future applications of LMEnt and limitations.

8.1 Future Applications of LMEnt

Knowledge plasticity LMEnt can be used to study the plasticity of knowledge in language models: identifying steps during pretraining when models are more receptive to acquiring new knowledge. This follows prior findings that pretrained models struggle to learn new facts (Lyle et al., 2022), excessive pretraining can make models resistant to fine-tuning (Springer et al., 2025), and LMs store a finite amount of parametric knowledge (Allen-Zhu and Li, 2025).

Improving factuality of LMs Since LMEnt provides entity mentions for each chunk in the pretraining corpus, a possible extension is to leverage them to improve the model's factuality and mitigate hallucinations. For example, one could experiment with different data ordering methods, like grouping together all the chunks mentioning the same entity, or augmenting the pretraining data by replacing implicit mentions with explicit entity names.

Effect of other data sources In this work, we train models on a relatively small and knowledge-rich corpus, which results in LMs that are capable in knowledge tasks, yet perform poorly on out-of-distribution tasks, such as commonsense reasoning (§A.4). Modern LMs are trained on much larger corpora, often surpassing 10T tokens (Allal et al., 2025), derived from sources that are *knowledge-poor*, e.g., coding (Roziere et al., 2023; Hui et al., 2024), synthetic stories (Eldan and Li, 2023), and formal languages (Hu et al., 2025). The LMEnt pretraining dataset can be extended with such sources to analyze if the addition of these tokens improves factuality. The annotation pipeline can also be applied to other knowledge-rich sources that lack hyperlinks, since relying just on entity linking is still effective (`LMEnt -H -C`, Fig. 5).

Mechanistic interpretability The enhanced visibility that LMEnt provides into the training process facilitates a controlled, yet natural, setup for studying the formation of latent knowledge representations and circuits in LMs.

8.2 Limitations

Pretraining data, model size and architecture LMEnt presents a relatively small pretraining corpus and collection of compute-efficient models. Such experimental settings have been successful in recent years, enabling academia to develop procedures that were later scaled effectively, e.g., (Rafailov et al., 2023). As mentioned in §8.1, an interesting future direction is to scale LMEnt annotations to other sources—thereby increasing the size of the pretraining corpus and facilitating studies into how different types of data affect knowledge acquisition. Also, LMEnt can be easily extended to architectures beyond dense transformers like mixture-of-experts (Dai et al., 2024; Muennighoff et al., 2025; Cai et al., 2025), which is already supported by the OLMo framework (Groeneveld et al., 2024; Muennighoff et al., 2025).

Beyond pretraining Since models see the vast majority of tokens during pretraining, this stage is critical for understanding how they acquire and represent knowledge. While our study focuses on pretraining, it is typically followed by mid- and post-training phases. In these stages, models are presented with additional high-quality next-token prediction data and fine-tuning examples (OLMo et al., 2024; Kumar et al., 2025). Future work could extend LMEnt by adding annotations to the data used for mid- and post-training, and further train LMEnt models on this data.

Dense retrievers In this work, we do not compare LMEnt retrieval against dense retrievers such as DPR (Karpukhin et al., 2020) and ColBERT (Khattab and Zaharia, 2020), since sparse retrieval methods like WIMBD (Elazar et al., 2024) and Infinigram (Liu et al., 2024a) are the predominant approaches for retrieving relevant information in pretraining corpora.

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A Appendix: Supplementary Results

A.1 Pretraining Data Statistics

Tab. 2 summarizes statistics over the LMEnt pretraining corpus. These results support the overview provided in §4.

# tokens	# chunks	# entities	# batches	# mentions	# hyperlinks	# entities by H					# entity linking	# coref. mentions	# coref. clusters
			(steps)	total	(H)	1	(1, 10]	(10, 100]	(100, 1K]	> 1K			
3.6B	10.5M	7.3M	109K	400M	115M	993K	2.2M	967K	141K	13K	203M	151M	310M

Table 2: Statistics of the LMEnt pretraining corpus.

A.2 PopQA and PAQ Performance

In this section we present additional results for PopQA and PAQ accuracies as functions of compute budget. Fig. 12 shows the results for all questions and the subset of questions whose subject and answer entities co-occur moderately in PopQA, and Fig. 13 shows the results for all questions and the subset of questions whose subject and answer entities co-occur frequently in PAQ. These results support §5.1.

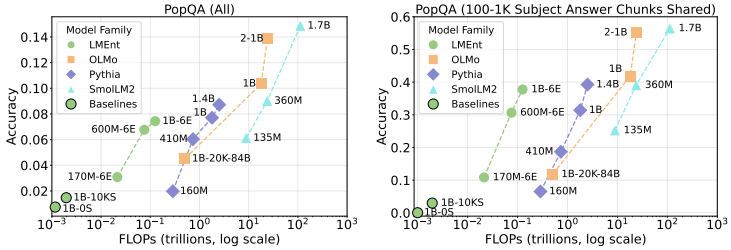


Figure 12: Accuracy on PopQA as a function of compute budget: (left) all questions and (right) questions for which the subject and answer entities appear together in 100-1K chunks (moderate).

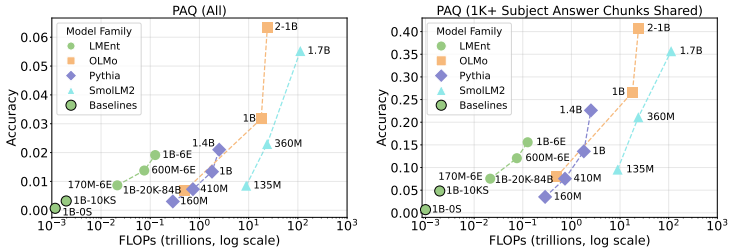


Figure 13: Accuracy on PAQ as a function of compute budget: (left) all entities and (right) questions for which the subject and answer entities appear together in 1K+ chunks (frequent).

A.3 Popularity and Fact Frequency Indicators for Model Behavior on Additional Models

In this section, we present additional results for PopQA accuracies sliced by popularity and fact frequency indicators. Fig. 14 shows the results for OLMo-2-1B (OLMo et al., 2024), Pythia-1.4B (Biderman et al., 2023), and SmolLM-2-1.7B (Allal et al., 2025). These results support §5.2.

A.4 Results on Additional Tasks

We report results on non-knowledge tasks in Tab. 3, which complement §5.1.

A.5 Empirical Experiment for Choosing LMEnt Score Thresholds

In this section, we describe the empirical experiment that determined the score thresholds for LMEnt retrieval (§5.2). To determine thresholds for the hyperlink score **H**, entity-linking score **EL**, and coreference scores **C** and **CC**, we sample 60 entities from PopQA based on hyperlinks counts. For each entity, we retrieve their scores using a fixed hyperlink threshold of $\mathbf{H} = 1$, while varying **EL**, **C**, and **CC** across the set $\{0.4, 0.5, 0.6, 0.7, 0.8\}$, and evaluate precision over chunks retrieved at different depths k. If $k > 100$, we randomly sample 100 of the k returned chunks and submit them to Gemini 2.5 Flash Preview 6-17 using the prompt shown in Fig. 20 to compute precision at k. Results are reported in Tab. 4.

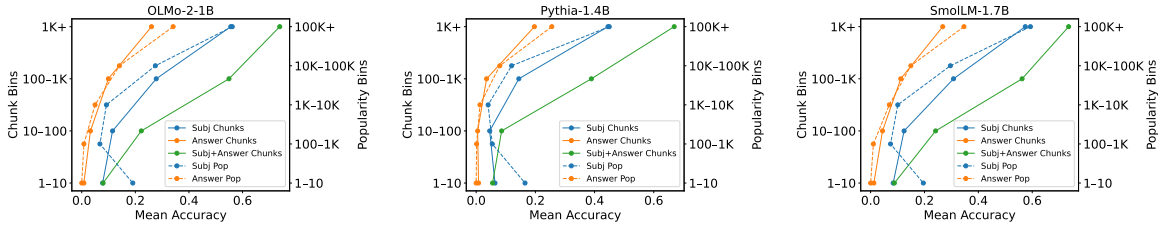


Figure 14: Accuracy of OLMo-2-1B (OLMo et al., 2024), Pythia-1.4B (Biderman et al., 2023), and SmolLM-2-1.7B (Allal et al., 2025) on PopQA, sliced by various indicators of popularity and fact frequency on PopQA.

Table 3: Performance of LMEnt models and baseline models on commonsense reasoning, multiple choice, and reading comprehension tasks.

Model	arc_challenge	arc_easy	boolq	copa	coqa	drop	gsm8k	hellaswag	jeopardy	naturalqs_open	openbookqa	piqa	sciq	squad	triviaqa	truthfulqa	winogrande
LMEnt-170M-1E	0.217	0.370	0.574	0.570	0.072	0.022	0.020	0.271	0.006	0.002	0.304	0.541	0.633	0.038	0.000	0.471	0.493
LMEnt-170M-2E	0.247	0.368	0.620	0.610	0.081	0.024	0.025	0.274	0.006	0.008	0.320	0.560	0.697	0.043	0.002	0.471	0.505
LMEnt-170M-4E	0.254	0.282	0.367	0.510	0.000	0.000	0.000	0.247	0.000	0.000	0.264	0.487	0.237	0.000	0.000	0.488	0.488
LMEnt-170M-6E	0.271	0.389	0.400	0.640	0.082	0.034	0.025	0.280	0.006	0.007	0.356	0.541	0.711	0.056	0.004	0.471	0.501
LMEnt-600M-1E	0.271	0.409	0.604	0.650	0.085	0.012	0.025	0.285	0.008	0.001	0.340	0.555	0.722	0.044	0.001	0.472	0.490
LMEnt-600M-2E	0.301	0.405	0.620	0.630	0.113	0.046	0.055	0.296	0.009	0.020	0.350	0.554	0.744	0.085	0.005	0.468	0.516
LMEnt-600M-4E	0.254	0.389	0.565	0.610	0.150	0.055	0.015	0.306	0.034	0.010	0.338	0.559	0.735	0.092	0.016	0.470	0.515
LMEnt-600M-6E	0.274	0.409	0.625	0.630	0.167	0.075	0.050	0.318	0.023	0.029	0.374	0.569	0.762	0.119	0.045	0.444	0.510
LMEnt-1B-1E	0.254	0.421	0.603	0.650	0.109	0.063	0.035	0.295	0.009	0.019	0.358	0.557	0.714	0.084	0.006	0.469	0.506
LMEnt-1B-2E	0.244	0.377	0.626	0.630	0.132	0.068	0.045	0.309	0.020	0.018	0.360	0.557	0.765	0.089	0.047	0.448	0.523
LMEnt-1B-4E	0.278	0.391	0.545	0.670	0.160	0.067	0.010	0.322	0.021	0.024	0.374	0.557	0.763	0.092	0.008	0.455	0.517
LMEnt-1B-6E	0.264	0.446	0.611	0.680	0.157	0.070	0.035	0.328	0.043	0.031	0.390	0.555	0.770	0.107	0.025	0.439	0.529
OLMo-1B	0.338	0.565	0.617	0.770	0.574	0.218	0.030	0.627	0.342	0.123	0.434	0.729	0.879	0.624	0.324	0.329	0.594
OLMo-1B-20K-84B	0.288	0.458	0.640	0.760	0.269	0.079	0.000	0.433	0.059	0.051	0.406	0.676	0.790	0.251	0.085	0.394	0.531
OLMo-2-1B	0.435	0.635	0.646	0.800	0.694	0.353	0.395	0.688	0.641	0.191	0.474	0.746	0.953	0.818	0.512	0.369	0.654
Pythia-1.4B	0.348	0.539	0.582	0.760	0.566	0.206	0.020	0.540	0.273	0.098	0.434	0.707	0.874	0.565	0.250	0.386	0.566
Pythia-1B	0.331	0.525	0.623	0.740	0.519	0.185	0.025	0.478	0.195	0.077	0.378	0.683	0.890	0.534	0.188	0.389	0.533
Pythia-410M	0.301	0.467	0.588	0.700	0.428	0.166	0.040	0.393	0.069	0.058	0.360	0.663	0.833	0.438	0.104	0.410	0.532
Pythia-160M	0.311	0.423	0.498	0.640	0.218	0.136	0.020	0.300	0.019	0.031	0.324	0.606	0.746	0.176	0.042	0.442	0.502
SmolLM2-1.7B	0.468	0.704	0.733	0.840	0.720	0.259	0.315	0.695	0.706	0.194	0.496	0.760	0.940	0.771	0.545	0.367	0.671
SmolLM2-360M	0.448	0.649	0.620	0.760	0.596	0.190	0.045	0.551	0.516	0.107	0.444	0.710	0.905	0.656	0.305	0.334	0.591
SmolLM2-135M	0.365	0.558	0.617	0.660	0.429	0.135	0.025	0.410	0.244	0.079	0.422	0.670	0.842	0.467	0.175	0.388	0.530

A.6 Win Rates

In Fig. 15, we extend the experiments in (§5.2, Fig. 5) to compare win rates between the best string based method, CS-SS Canonical, and all other string-based variants (left), as well as CI-SS Canonical versus CI-SS Expanded (right). These results complement Fig. 5.

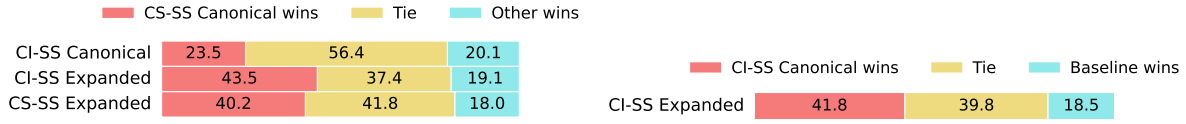


Figure 15: Pairwise win rates between CS-SS Canonical and other string-based methods (left). Pairwise win rates between CI-SS Canonical and CI-SS Expanded (right).

A.7 LMEnt provides better coverage for rare entities

Fig. 16 shows the distribution over the total number of retrieved chunks per entity. We observe that LMEnt offers greater coverage for tail and torso entities compared to Canonical string-based variants. This is especially significant since tail and torso entities together make up 99.7% of the total entities in the corpus (§A.1, Tab. 2). While it appears that Expanded string-based variants retrieve more chunks, this is likely noise due to alias ambiguity. There is also a noticeable difference in variance—specifically string-based methods show egregious retrieval failures, sometimes retrieving less than 10 documents for head entities and no documents for torso entities. These results support §5.2.

A.8 LMEnt wins across all entity frequency levels, often by large margins

Fig. 17 shows "Yes" chunk count differences between methods and the cumulative percentage of entities where one method outperforms the other, across entities of different frequencies. For tail entities (right), LMEnt outperforms by a few chunks, but this is meaningful given the scarcity of mentions. For torso

Table 4: Empirical evaluation of LMEnt retrieval precision across multiple thresholds and retrieval depths k on the development set of 60 PopQA entities.

Score Thresholds	1	5	10	100	1K	10K	100K
H=1, EL = C = CC = 0.4	0.95	0.95	0.96	0.98	0.99	0.98	0.94
H=1, EL = C = CC = 0.5	0.95	0.95	0.96	0.98	0.99	0.99	0.91
H=1, EL = C = CC = 0.6	0.95	0.96	0.96	0.98	0.99	0.98	0.94
H=1, EL = C = CC = 0.7	0.95	0.96	0.96	0.98	0.99	0.98	0.91
H=1, EL = C = CC = 0.8	0.95	0.96	0.96	0.98	0.97	0.88	0.96

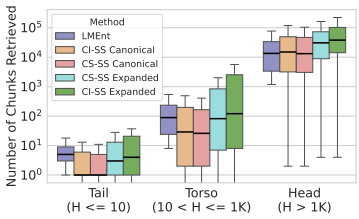


Figure 16: Distribution of the number of chunks retrieved by LMEnt, and both Canonical and Expanded variants of CI−SS and CS−SS. Retrieval was performed for 1K PopQA entities, selected via stratified sampling based on hyperlink counts. LMEnt retrieves more chunks than the Canonical variants for torso and tail entities, which together account for 99.7% of all entities in Wikipedia. The higher number of chunks returned for head entities by Expanded variants is likely bloated by noisy retrieval (Fig. 6).

entities middle), LMEnt wins by a substantial margin ($\geq$ 20 additional correct chunks) in 40% of cases. For head entities (left), the margin is typically smaller (1–25 chunks), suggesting that string-based search performs more competitively in these cases – though, LMEnt still outperforms it on $\geq$ 60% of entities.

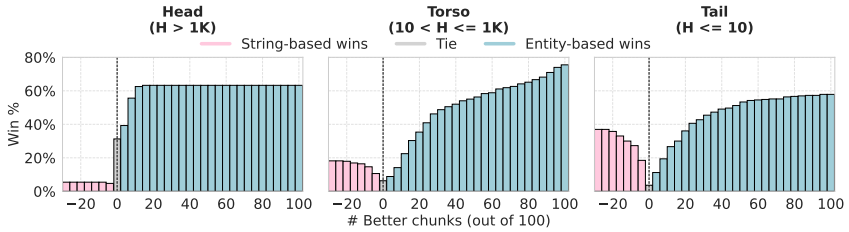


Figure 17: Comparison between LMEnt retrieval and the best performing string-based approach (CS−SS Canonical). Results are grouped by frequency of hyperlinks; tail entities (right), torso entities (center), and head entities (left).

A.9 Knowledge Acquisition during Pretraining

Fig. 18 shows the net gains in facts learned between LMEnt-1B-6E checkpoints across frequency bins of subject and answer entity co-occurence, and additional examples of three specific relations. The split of % of facts learned and forgotten is in §6, Fig. 8.

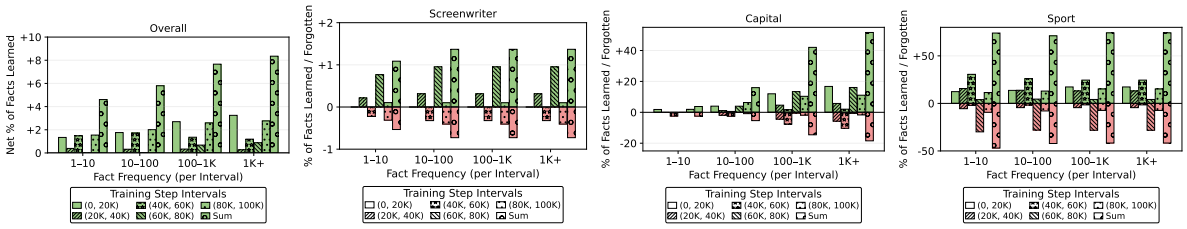


Figure 18: Percentage of net gains in facts learned between intermediate checkpoints of LMEnt-1B-6E (left). Additional examples of the relation-specific analysis in in §6.

B Appendix: Additional Implementation Details

B.1 Running Maverick At Scale

The backbone of Maverick (Martinelli et al., 2024) is DeBERTa (He et al., 2021) which has quadratic attention complexity with respect to sequence length. Therefore, running Maverick on long documents is impractical, e.g. 15K token long documents take 40-50GB VRAM. Therefore, we reduce the memory footprint, and accelerate and parallelize inference across multiple documents by replacing the DeBERTa backbone with a FlashAttention-based implementation (Knowledgator, 2025). Then, we apply Maverick to non-overlapping windows of 50K tokens, which may result in multiple distinct coreference clusters for the same entity. In these cases, we rely on the coverage of hyperlinks and the entity linker (which runs on the entire document) to map each cluster to the target entity's QID.

B.2 Coreference scoring

In this section we detail how we compute the score derived from coreference resolution, and how we resolve ambiguity when multiple entities are associated with a single coreference cluster (§2.2).

Coref (C) is designed to associate shortened references, not previously found by hyperlinks or entity linking, with their canonical names, e.g. "*the hospital*" with "*John R. Oishei Children's Hospital*". The coreference cluster of "*the hospital*" contains "*the hospital*" and "*John R. Oishei Children's Hospital in* Buffalo". There are two entities related to this cluster, with QIDs: "Q93565992" and "Q40435". We can reduce ambiguity by leveraging the shared substring of "*(H,h)ospital*" to promote "*the hospital*" being mapped to "Q93565992" more strongly. We represent this promotion by a score denoted by **C**, which we formally define below.

Let the coreference cluster associated with m be $\mathcal{C}$, and define the sets of hyperlink and entity linking mentions that overlap with $\mathcal{C}$ as $\mathcal{C}_{\text{H}}$ and $\mathcal{C}_{\text{EL}}$, respectively. To calculate $\mathbf{C}(m, e)$, for each mention $m' \in \mathcal{C}_{\text{H}} \cup \mathcal{C}_{\text{EL}}$, we compute the longest common substring (LCS) with m. Let $\text{sim}(m, m')$ be the harmonic mean of the LCS overlap ratios between mention m and m', defined as:

$$\text{sim}(m, m') = \begin{cases} 0.0 & \text{if } |m| = 0 \text{ or } |m'| = 0 \\ \frac{2 \cdot a \cdot b}{a+b} & \text{otherwise} \end{cases} \quad \text{where } a = \frac{\text{LCS}(m, m')}{|m|},\ b = \frac{\text{LCS}(m, m')}{|m'|} \tag{1}$$

We weight either the hyperlink or entity linking score of m' for e by its textual similarity to m:

$$\text{support}(m, m', e) = \text{sim}(m, m') \cdot \begin{cases} \mathbf{H}(m', e) & \text{if } m' \in \mathcal{C}_{\text{H}} \\ \mathbf{EL}(m', e) & \text{if } m' \in \mathcal{C}_{\text{EL}} \end{cases} \tag{2}$$

Finally, the **C** score for mention m and entity e is given by the maximum support score over all other mentions $m' \in \mathcal{C}$:

$$\mathbf{C}(m, e) = \max_{m' \in \mathcal{C}_{\text{H}} \cup \mathcal{C}_{\text{EL}}} \text{support}(m, m', e) \tag{3}$$

Coref-Cluster (CC) connects generic indirect mentions (that don't share textual similarity with the entity name) to their canonical entity names, e.g. "*their*" with "*Buffalo Bills*". In this case, we compute a distribution of scores over all the entities linked to some mention in the cluster. The score $\mathbf{CC}(e)$ is defined per entity per cluster, and is shared across all mentions $m \in \mathcal{C}$.

For each mention $m \in \mathcal{C}$ and entity e, we look at the existing source scores and compute a $\text{m}_{\text{support}}$ score. We don't aggregate the existing source scores (e.g. compute an average), because we found that this disproportionately rewarded single mentions identified by multiple sources and outweighed multiple mentions found by single sources, leading to incorrect cluster to entity associations. The $\mathcal{C}_{\text{support}}$ for entity e in cluster $\mathcal{C}$ is simply the sum over all $\text{m}_{\text{support}}$ scores.

$$\mathrm{m}_{\text{support}}(m,e) = \begin{cases} \mathbf{H}(m,e) & \text{if } m \in \mathcal{C}_{\text{H}} \\ \mathbf{EL}(m,e) & \text{if } m \in \mathcal{C}_{\text{EL}} \\ \mathbf{C}(m,e) & \text{otherwise} \end{cases} \quad \mathcal{C}_{\text{support}}(e) = \sum_{m \in \mathcal{C}} \mathrm{m}_{\text{support}}(m,e) \tag{4}$$

To compute a distribution of scores over all entities mentioned in the cluster, we compute a softmax over $\mathbf{CC}(\mathcal{C}, e')$ scores for the set of entities e' supported by at least one mention in $\mathcal{C}$, denoted by $\mathcal{E}_{\mathcal{C}}$.

$$\mathbf{CC}(\mathcal{C}, e) = \frac{\exp\left(\mathcal{C}_{\text{support}}(e)\right)}{\sum_{e' \in \mathcal{E}_{\mathcal{C}}} \exp\left(\mathcal{C}_{\text{support}}(e')\right)} \tag{5}$$

B.3 Error Analysis of `LMEnt` Annotations

This section describes a qualitative error analysis in which we sampled 112 mentions retrieved using `LMEnt`, and manually analyzed the annotation errors observed and their frequency. We saw three errors total (2.7%) that are displayed in Fig. 19. The errors were results of entity linking failures and one coreference resolution failure.

entity	entity_description	qid	mention	title	highlighted_text	scores_by_source
Face to Face (1976 film)	Face to Face (1976 film) - 1976 film by Ingmar Bergman	Q388846	Ansikte mot ansikte	Massimo Di Gesu	Di Gesu's works such as "<START>Ansikte mot ansikte<END>" (for Serate Musicali - Milano, and Ente Concerti Pesaro) and "WOLFiliGrANG" (at the Rovereto Mozar	{'coref': 0.0, 'coref_cluster': 0.0, 'entity_linking': 0.96, 'hyperlinks': 0.0}
The Message (short story)	The Message (short story) - science fiction short story	Q7751133	George Kilroy	Jack Donahue	ty lashes as punishment. Donahue escaped to the bush from the Quakers Hill farm with two men named <START>George Kilroy<END> and William Smith. They formed an outlaw gang known as "The Strippers," since they stripped wealthy	{'coref': 0.91, 'coref_cluster': 0.6, 'entity_linking': 0.91, 'hyperlinks': 0.0}
Happy End (musical)	Happy End (musical) - 1929 three-act musical comedy by Kurt Weill, Elisabeth Hauptmann, and Bertolt Brecht	Q1584256	March Ahead	Nagorik Shakti	rty, and featured addresses to women, the young, and the expatriates. "Bangladesh egiye cholo" or "<START>March Ahead<END>, Bangladesh" was to be the theme title of the movement, as declared in the second letter to the peo	{'coref': 0.0, 'coref_cluster': 0.0, 'entity_linking': 0.93, 'hyperlinks': 0.0}

Figure 19: Error analysis of `LMEnt` entity mentions.

B.4 Pretrained Models

Details of the `LMEnt` models are described here and support the overview in §4. The 170M, 600M, and 1B models use $(\text{layers}, \text{hidden dimension})$ configurations of $(10, 768)$, $(16, 1344)$, and $(16, 2048)$, respectively. A hyperparameter search was conducted over the following ranges: global batch size $\{16{,}384,\ 32{,}768,\ 65{,}536,\ 131{,}072,\ 262{,}144\}$, peak learning rate $\{3\times10^{-4},\ 6\times10^{-4},\ 8\times10^{-4},\ 1.2\times10^{-3},\ 3\times10^{-3},\ 5\times10^{-3}\}$, and weight decay $\{0.005,\ 0.05,\ 0.1\}$. Hyperparameters were selected based on the minimal final training perplexity. All `LMEnt` models were trained using the AdamW optimizer with a global batch size of $32{,}768$, rank batch size of $8{,}192$, peak learning rate of 5×10^{-3}, weight decay of 0.05, and $1{,}000$ warmup steps.

B.5 LLM Judge for Chunk Precision

To automatically evaluate the retrieval quality of both `LMEnt` (our entity-based method) and string-based baselines (see § 5.2), we use Gemini 2.5 Flash Preview 6-17. For each of the 1K entities in the test set, we apply `LMEnt`, its ablations, and string-based methods `CS-SS-(Canonical, Expanded)` and `CI-SS-(Canonical, Expanded)`, to retrieve sets of relevant chunks. For a retrieved set containing more than 100 chunks, we randomly sample 100 chunks from it for evaluation.

To determine whether a chunk mentions the entity, we prompt Gemini using the instruction shown in Fig. 20. To provide context, we include the entity's description from ReFinED (Ayoola et al., 2022), which consists of the Wikipedia article title and the first sentence in the article. For each chunk, we identify matching mentions of the entity to create a short context window for Gemini. For `LMEnt`, mentions are selected based on QID and score thresholds. For string-based methods, we use the `highlight` block in the Elasticsearch (Banon, 2010) query to extract character spans that match the entity name. Around each mention, we select a 130-character window to give Gemini some context of the mention in the chunk text. We evaluate up to three mentions per chunk. If Gemini returns "Yes" for any, the chunk is marked as mentioning the entity. If all are judged "No", the chunk is considered not to mention it. For `LMEnt`, mentions are prioritized by a weighted sum of their scores, and for string-based methods, mentions are evaluated in the order they appear in the document.

Justifying use of LLM-as-a-Judge To support our use of an LLM-as-a-Judge for evaluating model-generated responses, we use the alternative annotator test introduced by Calderon et al. (2025). This test determines whether the LLM's performance is comparable or better than that of a randomly chosen human annotator. In line with their methodology, we employed three human annotators (graduate students) and used a dataset of 100 entity mentions randomly sampled from 'Yes" and "No" judged chunks. For each mention, the annotators received the same inputs as the LLM judge: the entity name, entity description, and 130-character context window around the mention, and were asked to evaluate whether the chunk named the entity. Instructions are found in Fig. 21. Following Calderon et al. (2025), we set $\epsilon = 0.1$, which yielded a winning rate of $\omega = 1.00$ with a p-values of $0.001, 0.001$, and $8.28e^{-5}$ for all three annotators, indicating that the LLM can be relied on.

Goal: Determine if the provided Text directly mentions or discusses the specific Entity.

Process:

First, carefully read the Text and identify any specific names, terms, or phrases that appear and might relate to the Entity.
Second, compare these identified text mentions (if any) to the provided Entity.

Rules for "Directly Mentions or Discusses":
1. The Text must contain the exact name of the Entity or a clearly identifiable and standard part of the Entity's name (e.g., "Einstein" for "Albert Einstein") that contextually refers directly to the Entity.

2. The mention must refer to the Entity itself. Merely using the Entity's name as a descriptive modifier for something else associated with the Entity (e.g., "the Wahhabi school of Saudi Arabia" referring to the school, not the country) does not count as directly mentioning or discussing the Entity, unless the descriptive phrase is the Entity (e.g., "University of London" for the entity "University of London").

3. Exclusions (Do NOT count as direct mention/discussion):
Mentions of individuals or things related to the Entity but not the Entity itself (e.g., family members, associates, parts or products of the entity unless the mention clearly refers back to the entity).

4. Inferences based on context, related events, locations, or concepts if the Entity's name is not used to refer to the Entity itself according to Rule 3.

5. Distinguishing Similar Names: If the Text contains a name that is similar to the Entity's name but refers to a different person, place, or thing (e.g., a different historical figure, a different organization with a similar name, or differing spellings like "Elisabeth" vs. "Elizabeth" referring to different individuals), it does not count as a direct mention of the target Entity. The context must clearly indicate the reference is to the specific Entity provided.

6. Possessives, actions, or attributes must be logically consistent with the entity's type. Implausible associations, like assigning a teacher to Ennis, a town in County Clare, Ireland (e.g., "Ennis's teacher"), are invalid.

Consider each rule individually and assign an intermediate judgment of Pass or Fail. If any rule receives a Fail, the final decision must be No.

Output: Provide only the answer 'Yes' or 'No' based on the above rules. Do not include any explanation or other text.

Entity: **{entity_description}**
Text: **{text}**
Answer:

Figure 20: Prompt given to Gemini to automatically judge whether a chunk mentions an entity directly.

B.6 Converting Queries to Cloze-Style Prompts

Since `LMEnt` models are not instruction-tuned, we evaluate them on PAQ by converting each question into a cloze-style prompt, where the expected answer is the next predicted phrase. To perform this transformation, we use Gemini 2.5 Flash Preview 6-17 with the prompt shown in Fig. 22.

Instructions

In this task, you will be given an Entity, a short Description of the Entity, and a short piece of Text that may or may not mention the Entity.

Your goal is to determine if the provided Text directly mentions or discusses the specific Entity.

To complete the task, perform the following steps:

(1) Carefully read the Text and identify any specific names, terms, or phrases that appear and might relate to the Entity.
(2) Compare these identified text mentions (if any) to the provided Entity according to the rules below. Consider each rule individually and assign an intermediate judgment of Pass or Fail per rule.
(3) Indicate your final decision as either 'Yes' or 'No':
    - If any rule receives a Fail, the final decision must be No.
    - Otherwise (if all rules received a Pass), the final decision should be Yes.

Rules for "Directly Mentions or Discusses":

(1) The text must contain the exact name of the Entity or a clearly identifiable and standard part of the Entity's name (e.g., "Einstein" for "Albert Einstein") that contextually refers directly to the Entity.

(2) The mention must refer to the Entity itself. Merely using the Entity's name as a descriptive modifier for something else associated with the Entity (e.g., "the Wahhabi school of Saudi Arabia" referring to the school, not the country) does not count as directly mentioning or discussing the Entity Saudi Arabia.

(3) Make inferences based on context, related events, locations, or concepts if the Entity's name is not used to refer to the Entity itself according to Rules 1, 2.

Exclusions (Do NOT count as direct mention/discussion):

(1) Mentions of individuals or things related to the Entity but not the Entity itself (e.g., family members, associates, parts or products of the entity unless the mention clearly refers back to the entity).

(2) Distinguishing Similar Names: If the Text contains a name that is similar to the Entity's name but refers to a different person, place, or thing (e.g., a different historical figure, a different organization with a similar name, or differing spellings like "Elisabeth" vs. "Elizabeth" referring to different individuals), it does not count as a direct mention of the target Entity.

(3) Possessives, actions, or attributes must be logically consistent with the entity's type. Implausible associations, like assigning a teacher to *Ennis, a town in County Clare, Ireland* (e.g., "Ennis's teacher"), are invalid.

Figure 21: Instructions given to annotators for evaluating LLM-as-a-judge.

Your goal is to rephrase the input Question as a Cloze Statement so that the Answer is the next word that would logically complete the sentence.

Instructions:
1. Do not lose any context from the question. The cloze statement must preserve the full meaning and specificity of the original question.
For example,
Question: What TV network does Funny or Die Presents air on?
Answer: HBO
Cloze Statement: Funny or Die Presents airs on the TV network

2. Rephrase the question into a natural, grammatically correct sentence. If the question is incomplete or unclear, you may add minimal language to clarify the intended answer.
For example,
Question: stevie cameron worked as a confidential informant for the rcmp during which controversial
Cloze Statement: Stevie Cameron worked as a confidential informant for the rcmp during the controversial

3. End the statement right before the answer.

4. Use any specific labels or terms from the question. For example, if the question specifies "capital city" or "TV network," include that exact phrase in the cloze.

Output ONLY the Cloze Statement at the end of thinking.

Question: **{question}**
Answer: **{answer}**
Cloze Statement:

Figure 22: Prompt given to Gemini 2.5 Flash Preview 6-17 to convert PAQ questions to cloze-style prompts.